

Math 455 notes

Shan Gao

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Personal notes for math 455

Remarks 1. We begin with some general remarks about Hilbert spaces.

First of all if $\{e_i\}_{i=1}^{\infty}$ is orthonormal, then

$$\|f\|_2^2 = \sum_{i=1}^{\infty} \langle f, e_i \rangle^2$$

Furthermore, TFAE:

- $\{e_i\}_{i=1}^{\infty}$ forms a basis of H
- if $\langle f, e_i \rangle = 0 \forall i$, then $f = 0$
- Let $a_i = \langle f, e_i \rangle$ and $S_n = \sum_{i=1}^n a_i e_i$, then $\|f - S_n(f)\|_2^2 \rightarrow 0$ as $n \rightarrow \infty$
- Defining

$$f \rightarrow \sum_{i=1}^n \langle f, e_i \rangle e_i = F_N(f)$$

as the projection of f onto the space $\{e_1, \dots, e_n\}$, then $\forall f \in H$,

$$\|f\|_2^2 = \sum_{i=1}^{\infty} |\langle f, e_i \rangle|^2 = \sum_{i=1}^{\infty} |a_i|^2$$

Also finite linear combinations of $\{e_i\}_{i=1}^{\infty}$ are dense in H .

Remark 2. A fact that will be useful for the homework is that given

$f \in L^1(-\pi, \pi)$, then the n -th Fourier coefficient of f is $\hat{f}(n) = \langle f(x), e^{inx} \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$ where $n \in \mathbb{Z}$.

Theorem 3 (Hilbert space basis existence). *Every Hilbert space has an orthonormal basis*

The proof for this uses Zorn's lemma (axiom of choice)

Lemma 4 (Zorns). *Every non-empty partially ordered set in which every chain is bounded also contains a maximal element*

Definition 5 (Partially ordered set). $(X, \leq)$ is a partially ordered set if

- $x \leq x \forall x \in X$
- if $x, y \in X$ with $x \leq y$ and $y \leq x$, then $x = y$
- if $x, y, z \in X$, with $x \leq y$, $y \leq z$, then $x \leq z$

Definition 6 (Chain). A chain A is a totally ordered subset of X where $\forall x, y \in A$, either $x \leq y$ or $y \leq x$.

Definition 7 (Bounded chain). $Y \subseteq X$ is bounded if $\exists z \in X$ such that $\forall y \in Y, y \leq z$.

Definition 8 (maximal element). A maximal element m of X is an element where if $m \leq x$ for $x \in X$, then $x = m$.

We want to use Zorn's lemma to prove

Proof of Theorem 3. Let H be a Hilbert space, and let $\mathcal{O}$ be the collection of all orthonormal subsets of H . $\mathcal{O}$ is non empty because $\{x\} \in \mathcal{O}$ where x is just any element of H with $\|x\| = 1$. We can define a partial ordering by set inclusion. Observing that every chain $\mathcal{A} \subseteq \mathcal{O}$ has an upper bound, by taking the union of every set in $\mathcal{A}$. Knowing this Zorn's lemma guarantees the existence of a maximal element, call it B , we want to show that this forms a basis of H . It's enough, as shown previously, to show that the linear span of B is dense in H , and this is if and only if the closure of $\text{span}B$, denoted, $\overline{\text{span}B}$ ends up being the whole space.

Remark 9 (Orthogonal decomposition theorem). Letting H be a hilbert space and M be a closed subset of H , define the orthogonal complement of M as

$$M^\perp = \{y \in X : \langle x, y \rangle = 0, x \in X\}$$

then for any $x \in H$, $\exists!$ decomposition where

$$x = Px + Qx$$

where $Px \in M$ and $Qx \in M^\perp$, where P and Q are the projection of x to M and $M^\perp$, respectively. Generally we have that $H = M \oplus M^\perp$

Thus we can take the closure of the linear span of B , $\overline{\text{span}B}$, by the orthogonal decomposition theorem, if $\overline{\text{span}B} \neq H$, then

$$H = \overline{\text{span}B} + \overline{\text{span}B}^\perp$$

But that means exists some $y \in \overline{\text{span}B}^\perp$ with $\|y\| = 1$ that is orthonormal to every element in B , since $B \subseteq \overline{\text{span}B}$, thus $B \cup \{y\}$ is a larger orthonormal subset, contradicting maximality of B , and we are done. $\square$

Definition 10 (Trigonometric Polynomials). Trigonometric polynomials are polynomials of the form

$$\sum_{n=-N}^{+N} a_n e^{inx} \quad (a_n \in \mathbb{C})$$

Proposition 11. Trigonometric polynomials are dense in $L^2([-\pi, \pi])$

Recalling that $C([- \pi, \pi])$ is dense in $L^2([- \pi, \pi])$. It suffices to show that for all $f \in C([- \pi, \pi])$, then $\forall \varepsilon > 0 \exists P$ a trig polynomial such that $\|f - P\|_2 < \varepsilon$. In fact we can show that $\exists P$ such that $\|f - P\|_\infty < \varepsilon$, which is even a stronger result.

Theorem 12 (Stone-Waierstrass). *Let X be a compact, Hausdorff topological space. Let $\mathcal{A}$ be an algebra of continuous functions on X that separates points and contains constant functions. Then $\mathcal{A}$ is dense in $C(X)$.*

Before any proofs, we define an algebra

Definition 13 (Algebra (not set theoretic)). define $\mathcal{A}$ to be an algebra, which is a set with multiplication addition and scalar multiplication on a field $\mathbb{K}$ such that $f, g \in \mathcal{A}$, and $c_1, c_2 \in \mathbb{K} \implies c_1 f + c_2 g \in \mathcal{A}$, and $f \cdot g \in \mathcal{A}$. (Examples include polynomials, trigonometric polynomials e^{inx})

Definition 14 (Separates points). We say a collection $\mathcal{F}$ of functions on X separates points if $\forall x, y \in X, x \neq y, \exists f \in \mathcal{F}$ such that $f(x) \neq f(y)$.

We ain't gonna give a full proof, but for now consider a more restricted result.

Theorem 15. $\forall f \in C([- \pi, \pi]), \forall \varepsilon > 0, \exists P$ a trigonometric polynomial such that $\|f - P\|_\infty < \varepsilon$.

Proof. Let $Q_1(x), Q_2(x), \dots$ be trigonometric polynomials such that

- $Q_k \geq 0 \forall k \in \mathbb{N} \forall x \in \mathbb{R}$
- $\frac{1}{2\pi} \int_{-\pi}^{\pi} Q_k(x) dx = 1 \forall k \in \mathbb{N}$
- Letting $\eta_k(\delta) = \sup_{\delta \leq |x| \leq \pi} Q_k(x)$. Then $\lim_{k \rightarrow \infty} \eta_k(\delta) = 0$

The higher level concept of the proof, is that given a sequence of trigonometric polynomials with these properties, given any function $f \in C([- \pi, \pi])$, we can construct a function, that is also a trigonometric polynomial such that for all $y \in [- \pi, \pi]$, these two are arbitrarily close. Notice that defining

$$P_k(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-y) Q_k(y) dy$$

. Using a change of variables and using the periodicity of e^{inx} we get that the latter integral is equal to

$$\begin{aligned} P_k(x) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) Q_k(x-y) dy = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) \sum_{n=-N}^{+N} a_{n,k} e^{in(x-y)} dy \\ &= \sum_{n=-N}^{+N} a_{n,k} e^{inx} \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) e^{-iny} dy \\ &= \sum_{n=-N}^{+N} a_{n,k} \hat{f}_n(y) e^{inx} \end{aligned}$$

which it itself a trigonometric polynomial, and then we take any function $f \in C([- \pi, \pi])$. Knowing this, let $\varepsilon > 0$, since f is continuous on a compact interval, then it's uniform continuous, meaning that $\forall x_1, x_2 \in [- \pi, \pi]$ if $|x_1 - x_2| < \delta$, then $|f(x_1) - f(x_2)| < \varepsilon$ and then we just need to evaluate

$$|f(y) - P_k(y)| = \frac{1}{2\pi} \int_{-\pi}^{\pi} (f(y-x) - f(y))Q_k(y)dx = I_1 + I_2$$

, since $\int Q_k(x) = 1$, we can bring $f(y)$ into the integral, since it doesn't depend on x . where

$$I_1 = \frac{1}{2\pi} \int_{-\delta}^{\delta} (f(y-x) - f(y))Q_k(y)dx$$

and

$$I_2 = \frac{1}{2\pi} \int_{[-\pi, -\delta] \cup [\delta, \pi]} (f(y-x) - f(y))Q_k(y)dx$$

Observing that

$$\begin{aligned} I_1 &= \frac{1}{2\pi} \int_{-\delta}^{\delta} (f(y-x) - f(y))Q_k(y)dx \\ &\leq \frac{1}{2\pi} \int_{-\delta}^{\delta} |f(y-x) - f(y)|Q_k(y)dx \\ &\leq \frac{1}{2\pi} \int_{-\delta}^{\delta} \varepsilon Q_k(y)dx \leq \varepsilon \end{aligned}$$

Since $|(y-x) - y| = |x| \leq \delta$ in the case of I_1 Now we just need to take care of I_2 . Note that $\exists M \in \mathbb{N}$ such that $|f(y-x) - f(y)| \leq \sup_{a,b \in [-\pi, \pi]} |f(b) - f(a)| \leq 2M$ since f reaches it's maxima on a compact interval.

$$\begin{aligned} I_2 &= \frac{1}{2\pi} \int_{[-\pi, -\delta] \cup [\delta, \pi]} (f(y-x) - f(y))Q_k(y)dx \\ &\leq \frac{1}{2\pi} \int_{[-\pi, -\delta] \cup [\delta, \pi]} 2M\eta_k(\delta)dx = 2M\eta_k(\delta) \end{aligned}$$

which vanishes to 0 as $k \rightarrow \infty$, by the third property, thus proving that the difference vanishes, so we are done with the main part of the proof. Now we are left to construct $Q_1(x), Q_2(x), \dots$ with the properties as described above. We can just choose $Q_k(x) = c_k \left(\frac{1 + \cos x}{2} \right)^k$, where c_k is a constant such that the integral turns

out to be 1. and also observe that, given c_k chosen appropriately

$$\begin{aligned}
 1 &= \frac{c_k}{2\pi} \int_{-\pi}^{\pi} \left(\frac{1 + \cos x}{2} \right)^k dx \\
 &= \frac{c_k}{\pi} \int_0^{\pi} \left(\frac{1 + \cos x}{2} \right)^k dx \quad \text{by symmetry} \\
 &> \frac{c_k}{\pi} \int_0^{\pi} \left(\frac{1 + \cos x}{2} \right)^k \sin x \, dx \\
 &= \frac{c_k}{n} \int_1^0 u^k (-2du) \\
 &= \frac{c_k}{n} \int_0^1 u^k (2du) \\
 &= \frac{2c_k}{\pi} \cdot \frac{1}{k+1}
 \end{aligned}$$

Using the substitution $u = \frac{1+\cos x}{2}$ and $du = -\frac{1}{2} \sin x dx$. We then get that $c_k < \frac{\pi}{2}(k+1)$ so $Q_k(x) < \frac{\pi}{2}(k+1) \left(\frac{1+\cos x}{2} \right)^k \rightarrow 0$ for $x \neq 0$, so we recover property 3. □

Corollary 16. *trigonometric polynomials are dense in $L^2([-\pi, \pi])$*

Proof. Trig polys are dense in $C([-\pi, \pi])$ under $\|\cdot\|_{\infty}$. If $|P_k(x) - f(x)| < \varepsilon$, then we simply get that

$$\|P_k - f\|_2 = \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} |f(x) - P_k(x)|^2 dx \right)^{\frac{1}{2}} < \sqrt{\varepsilon^2} = \varepsilon$$
□

Theorem 17. *Let X, Y be Banach spaces, $F: X \rightarrow Y$ bounded linear operator, and F is surjective onto Y . Let U be the open unit ball in X and V be the open unit ball. Then $\exists \delta > 0$ such that $F(U) \supset \delta V \implies F(B_r(x_0)) \supseteq B_{\delta r}(F(x_0))$*

Proof. $\forall y \in Y, \exists x \in X$, such that $F(x) = y$. If $\|x\| \leq k$, then we have something $y \in F(k \cdot U)$, $Y = \bigcup_{k \in \mathbb{N}} F(k \cdot U)$, Y complete. By Baire Category theorem, $\exists k$ such that $\overline{F(k \cdot U)} \supseteq W$, for some W non empty open set in Y . For every $y \in W, \exists \{x_i\}_i \subseteq k \cdot U$ such that $F(x_i) \rightarrow y$, this is by the fact that y in the closure of $F(k \cdot U)$ and the fact that F is onto. So let k be sufficiently large such that $\overline{F(k \cdot U)}$ contains non empty interior, so open set W . Let $y_0 \in W, \exists \rho > 0$ such that $B(y_0, \rho) \subseteq W$, then $y_0 + y \in W$, if $\|y\| < \rho$, then $\exists \{x'_i\}_i, \{x''_i\}_i \subseteq k \cdot U$ such that $F(x'_i) \rightarrow y_0, F(x''_i) \rightarrow y_0 + y, F(x''_i - x'_i) \rightarrow y_0 + y - y_0 = y$. □

Let $f \in L^1([-\Pi])$ where $\Pi = [-\pi, \pi]$. Recall Fourier coefficients of f are given by

$$\hat{f}(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$$

We can also define a partial sum of the Fourier convolution

$$(S_n f)(x) = \sum_{k=-n}^n \hat{f}(k) e^{ikx} dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \underbrace{\left(\sum_{k=-n}^n e^{ikx} \right)}_{\text{Dirichlet Kernel } D_n(x)} dx$$

References